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HOOD LATCH FOR VEHICLE

Abstract

A pop-up system (26) for a latch mechanism (24) of a vehicle, including: a main lever (30) pivotally mounted to a plate (32) of the latch mechanism (24); a split lever (36) pivotally mounted to the plate (32); a main spring (28) for providing a biasing force to the main lever (30) via the split lever (36); a bell crank (38) pivotally mounted to the main lever (30), the bell crank (38) being operably coupled to the split lever (36) at one end and a link (40) at another end, the link (40) pivotally mounted to a cinch lever (42), the cinch lever (42) being pivotally mounted to the plate (32); and a pawl (46) pivotally mounted to the plate (32), the pawl (46) being moveable between at least an engaged position wherein the pawl (46) maintains the split lever (36) in a first position wherein a biasing force of the main spring is prevented from being applied to the main lever (30) and a disengaged position wherein the biasing force of the main spring is applied to the main lever (30) in order to rotate the main lever into an open position when the latch mechanism (24) is released.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/557,343 filed on Feb. 23, 2024, the entire contents of which are incorporated herein by reference thereto.

BACKGROUND

[0002] Exemplary embodiments of the present disclosure pertain to the art of vehicle latches and more particularly, to a hood latch for a vehicle.

[0003] Vehicles can be configured to have a front storage compartment that is accessible through a hood that is capable of being manipulated between an open and closed position. In order to retain the hood in a latched or closed position, a latch is provided to maintain the hood in a latched or closed position. These front storage compartments may be referred to as a “Frunk” that has a latch or latch device for retaining the hood in the closed position. In some configurations, the latch or latch device is provided with a pop-up system that when actuated applies a spring biasing force to the hood in order to move it from the closed position to an open position. This spring biasing force is provided by a spring that must be compressed when the hood is being closed and/or latched. Accordingly, it is desirable to provide a latch or latch mechanism for a vehicle hood wherein the compression of the spring does not interfere with the latching process.

BRIEF DESCRIPTION

[0004] Disclosed is a pop-up system for a latch mechanism of a vehicle, including: a main lever pivotally mounted to a plate of the latch mechanism; a split lever pivotally mounted to the plate; a main spring for providing a biasing force to the main lever via the split lever; a bell crank pivotally mounted to the main lever, the bell crank being operably coupled to the split lever at one end and a link at another end, the link pivotally mounted to a cinch lever, the cinch lever being pivotally mounted to the plate; and a pawl pivotally mounted to the plate, the pawl being moveable between at least an engaged position wherein the pawl maintains the split lever in a first position wherein a biasing force of the main spring is prevented from being applied to the main lever and a disengaged position wherein the biasing force of the main spring is applied to the main lever in order to rotate the main lever into an open position when the latch mechanism is released.

[0005] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the split lever has a pin or protrusion that is configured to contact a leg of the main spring.

[0006] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the split lever has a tab portion configured to contact an end of the pawl when the pawl is in the engaged position.

[0007] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, a rotational force is applied to the main lever by an edge of the tab portion of the split lever contacting an edge portion of the main lever.

[0008] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the bell crank has a tab portion configured to contact a tab portion of the split lever when the split lever is in the first position.

[0009] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the main lever has a tab portion configured to contact the pawl and rotate the pawl into the disengaged position.

[0010] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the latch mechanism further comprises: a claw rotatably mounted to the plate and the cinch lever is operably coupled to the claw.

[0011] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the latch mechanism further comprises: a latch pawl rotatably mounted to the plate for movement between an engaged position and a disengaged position, the engaged position of the latch pawl retaining the claw in a closed position and the disengaged position of the latch pawl allowing the claw to rotate from the closed position to an open position.

[0012] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the split lever has a pin or protrusion that is configured contact a leg of the main spring.

[0013] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the split lever has a tab portion configured to contact an end of the pawl when the pawl is in the engaged position.

[0014] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, a rotational force is applied to the main lever by an edge of the tab portion of the split lever contacting an edge portion of the main lever.

[0015] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the bell crank has a tab portion configured to contact a tab portion of the split lever when the split lever is in the first position.

[0016] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the main lever has a tab portion configured to contact the pawl and rotate the pawl into the disengaged position.

[0017] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the latch mechanism is configured to retain a vehicle hood in a closed position by engaging a striker of the vehicle hood.

[0018] In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the latch mechanism is configured to retain a vehicle hood in a closed position by engaging a striker of the vehicle hood.

[0019] Also disclosed is a method for disengaging a main spring from a main lever of a pop-up system for a latch mechanism of a vehicle, including: pivotally mounting the main lever to a plate of the latch mechanism; pivotally mounting a split lever to the plate; providing a biasing force by a main spring to the main lever via the split lever; pivotally mounting a bell crank to the main lever, the bell crank being operably coupled to the split lever at one end and a link at another end, the link pivotally mounted to a cinch lever, the cinch lever being pivotally mounted to the plate; and pivotally mounting a pawl to the plate, the pawl being moveable between at least an engaged position wherein the pawl maintains the split lever in a first position wherein a biasing force of the main spring is prevented from being applied to the main lever and a disengaged position wherein the biasing force of the main spring is applied to the main lever in order rotate the main lever into an open position when the latch mechanism is released.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

[0021] FIG. 1 is a top view of a vehicle having a hood latch in accordance with the present disclosure;

[0022] FIGS. 2-4 are views illustrating a pop-up system for use with a latch or latch mechanism of

the present disclosure; and

[0023] FIGS. **5-8** are views illustrating a latch or latch mechanism with a pop-up system of the present disclosure.

DETAILED DESCRIPTION

[0024] A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

[0025] With initial reference to FIG. **1**, a motor vehicle is indicated generally at **10**. Motor vehicle **10** includes a body **12** that defines an occupant compartment **14**. Body **12** supports a prime mover **16** which may take on a variety of forms including internal combustion engines, hybrid engines, and electric motors. Prime mover **16** may be mounted in a rear portion (not separately labeled) of body **12** or may be arranged as a mid-engine configuration. In an embodiment, body **12** includes a forward storage compartment **18** having a hood **20**. As will be detailed herein, hood **20** includes a striker element that cooperates with a latch or latch mechanism **24** in order to retain the hood **20** in a closed or latched position and then subsequently release the hood from the closed or latched position so that it may be transitioned into an open position.

[0026] Referring now to FIGS. **1-4**, the latch or latch mechanism **24** in accordance with the present disclosure is illustrated. The latch or latch mechanism **24** includes a pop-up system or mechanism **26**. The pop-up system or mechanism **26** is configured to provide a pop-up force to the hood **20** when the latch or latch mechanism **24** is released. The pop-up force is provided by a main spring **28**. In accordance with various embodiments of the present disclosure, the pop-up system or mechanism **26** is configured to remove the an opposing biasing force of the main spring **28** when the latch or latch mechanism **24** is an open position so that components of the pop-up up system or mechanism **26** and/or the latch or latch mechanism **24** can transition from an open position to a closed position during a cinching operation without having to overcome the biasing force of the main spring **28**. In other words, the pop-up up system or mechanism **26** is configured to remove the biasing force of the main spring **28** first and prior to a portion of a cinching operation of the latch or latch mechanism **24** that transitions a claw of the latch or latch mechanism **24** into a closed position. Thereafter and when the cinching operation is complete and the latch or latch mechanism **24** closed (e.g., claw of the latch or latch mechanism **24** is in the closed position), the pop-up up system or mechanism **26** reintroduces the biasing force of the main spring **28** to the latch or latch mechanism **24** and/or the pop-up up system or mechanism **26** so that the biasing force of the main spring **28** can be used for popping open the hood **20** associated with the latch or latch mechanism **24** and/or the pop-up up system or mechanism **26**.

[0027] As mentioned above, a hood latch device that includes a pop-up mechanism or system allows the hood to be lifted and partially opened. So that the pop-up mechanism or system can lift the weight of the hood, it is necessary to provide the system with a strong enough spring. This spring must be manipulated or compressed during the closing action of the hood so that the compressed spring can be used to pop-up the hood once again. If the closing of the hood is carried out automatically through actuators. The actuators would have to overcome the opposition of the biasing force of the pop up spring as it is being manipulated or compressed. Therefore, it is desirable to be able to remove the opposing force of the pop-up spring and leave an unincumbered free path for the hood or frunk in its closing action and in particular when the closing action is being provided by a motorized cinching operation.

[0028] In accordance with the present disclosure, the pop-up up system or mechanism **26** has a lever that is split into two components that allows removable of the influence of the pop-up spring by means of a closure mechanism on the latch, in order to remove the influence of the main spring when the hood is in closing action. Additionally, a retention mechanism is provided that includes a pawl component that keeps the spring disabled until the latch is closed. This retention mechanism is overridden or released when the latch mechanism reaches its fully closed position.

[0029] In FIG. **2**, components of the pop-up up system or mechanism **26** of the latch or latch

mechanism **24** in accordance with the present disclosure is illustrated. In FIGS. 2-4, other components (e.g., claw, pawl etc.) of the latch or latch mechanism **24** are not illustrated for convenience of illustrating the pop-up up system or mechanism **26**. In FIG. 2, the pop-up up system or mechanism **26** of the latch or latch mechanism **24** is in an open position (e.g., hood opened). [0030] The pop-up up system or mechanism **26** of the latch or latch mechanism **24** includes a main lever **30** pivotally mounted to a plate **32** of the latch or latch mechanism **24** via a pin or post **34**. The main lever **30** is configured to provide a biasing or opening force to a striker of the hood **20** as it rotates from a closed position to an open position. In addition, the main lever **30** is capable of rotation between an open position and a closed position. The biasing or opening force is provided by main spring **28**, which in one non-limiting embodiment may be a torsion spring. Also pivotally mounted to the pin **34** is a split lever **36**. The split lever **36** has a pin or protrusion **33** that is configured contact a leg **35** of the main spring **28** as the split lever **36** rotates about pin **34**. As such, the aforementioned lever that is split into two components may be referred to as the main lever **30** and the split lever **36**.

[0031] A bell crank **38** is pivotally mounted to the main lever **30** via a pin or rivet **39**. The bell crank **38** is operably coupled to the split lever **36** at one end and a link **40** at another end. The link **40** is also pivotally mounted to a cinch lever **42** pivotally mounted to the plate **32** by for example, a pin **44**. The link **40** is pivotally secured to the cinch lever **42** by for example a pin or rivet **43**. The pop-up system or mechanism **26** also includes a pawl **46** pivotally mounted to the plate **32** by for example, a pin **48**. The pawl **46** is moveable between at least a first or an engaged position (FIG. 3) where the pawl **46** maintains the split lever **36** in a first or a spring disengaged position such that main spring **28** is not applying a biasing force to the main lever **30** and a second or a spring engaged position (See at least FIG. 4) wherein the biasing force of the main spring **28** is indeed applied to the main lever **30** in order to assist and popping up the hood of the vehicle when the latch or latch mechanism **24** is released.

[0032] In FIG. 3, the cinch lever **42** has been moved in the direction of arrow **50** through actuation of a cable **52** secured to the cinch lever **42**. The movement of cable **52** in the direction of arrow **50** may be facilitated by a motor **51** (illustrated schematically) or manual actuation. This movement may be referred to as a first movement of the cinch lever **42** in the direction of arrow **50**. This first movement of the cinch lever **42** in the direction of arrow **50** causes movement of the link **40** in the direction of arrow **52** which in turn rotates the bell crank **38** in the direction of arrow **54**. This movement of the bell crank **38** in the direction of arrow **54** causes rotational movement of the split lever **36** in the direction of arrow **56**, which in turn causes pin or protrusion **33** to engage leg **35** of the spring **28**. This movement of the bell crank **38** in the direction of arrow **54** is facilitated by tab portion **53** of the bell crank **38** contacting a complimentary tab portion **55** of the split lever **36** when the bell crank **38** rotates in the direction of arrow **54** which causes rotation of the split lever **36** in the direction of arrow **56** due to the contact of portions **53** and **55**.

[0033] As the split lever **36** is rotated in the direction of arrow **56** from the position illustrated in FIG. 2 to the position illustrated in FIG. 3, the pawl **46** is free to rotate generally in the direction of arrow **56** due to a biasing force applied to the pawl **46** by a spring **57** and a tab portion **58** of the split lever **36** is now engaged by an end **70** of the pawl **46** in order to maintain the split lever **36** in the first or spring disengaged position such that main spring **28** is not applying a biasing force to the main lever **30**. As such, the biasing force of the main spring **28** is not being applied to the main lever **30** and further movement of the cinch lever **42** in the direction of arrow **50** from the position illustrated in FIG. 3 to that illustrated in FIG. 4 does not have to overcome the biasing force of main spring **28**. Thus, less closing or cinching force is required. This movement of the cinch lever **42** in the direction of arrow **50** from the position illustrated in FIG. 3 to that illustrated in FIG. 4 may be referred to as a second movement of the cinch lever **42** in the direction of arrow **50**.

[0034] In addition, the main lever **30** is still in the open position and is ready to be moved into the closed position without being engaged to the main spring **28**. The movement of the main lever **30**

from the open position illustrated in FIG. 3 into the closed position illustrated in FIG. 4 will be facilitated by gravity in the direction of arrow 71 as the main lever 30 is now free to rotate in the direction of arrow 56. In other words, when the biasing force of main spring 28 is removed from the main lever 30 (e.g., movement of the split lever 36 from the position in FIG. 2 to FIG. 3), the main lever 30 is free to rotate by gravity from the position in FIG. 3 to the position illustrated in FIG. 4. If for some reason, the main lever 30 is stuck and does not rotate in the direction of arrow 56 from the position illustrated in FIG. 3 to that illustrated in FIG. 4, the movement of the cinch lever 42 in the direction of arrow 50 from the position illustrated in FIG. 3 to that illustrated in FIG. 4 (e.g., the second movement of the cinch lever 42 in the direction of arrow 50) also causes a claw of the latch or latch mechanism 24 to transition from its open position to a closed or latched position which causes a striker of the hood 20 to be captured or retained by the claw of the latch or latch mechanism 24. As the striker of the hood 20 is captured or retained by the claw of the latch or latch mechanism 24 the striker also contacts the main lever 30 and rotates the main lever 30 in the direction of arrow 56 from the position illustrated in FIG. 3 to that illustrated in FIG. 4.

[0035] As such, further movement of the cinch lever 42 in the direction of arrow 52 from the position illustrated in FIG. 3 to that of FIG. 4 will cause the main lever 30 to rotate in the direction of arrow 56 into the closed position of the main lever 30 illustrated in FIG. 4. This movement may be due to gravity in the direction of arrow 71 and/or a striker contacting the main lever 30 as a claw of the latch or latch mechanism 24 is rotated into a closed position thereby capturing and retaining the striker and causing it to contact the main lever 30 as the claw is rotated into the closed position and the striker rotates the main lever 30 in the direction of arrow 56.

[0036] The rotation of the claw of the latch or latch mechanism 24 into the closed position may be facilitated by a cinching operation of the latch or latch mechanism 24. The cinching operation of the latch or latch mechanism 24 causes the claw of the latch or latch mechanism 24 to rotate into a primary or latched position as the cinch lever 42 is operably coupled to the claw of the latch or latch mechanism 24.

[0037] Referring now to FIG. 4, the main lever 30 has been moved to the closed position and this movement (e.g., main lever 30 from the open position to the closed position whether by gravity and/or movement of the striker into the claw into the closed position of the claw and the latch or latch mechanism 24) causes a tab portion 72 of the main lever 30 to contact the pawl 46 and rotate it in the direction of arrow 54 such that end 70 of the pawl 46 no longer contacts tab portion 58 of the split lever 36. As such, a spring biasing force of the main spring 28 in the direction of arrow 74 is applied to pin or protrusion 33 of the split lever 36 which in turn causes an edge 76 of tab portion 58 of the split lever 36 to provide a biasing force to an edge portion 78 of the main lever 30 which in turn provides a rotational force in the direction of arrow 54 to the main lever 30. As such, and as illustrated in at least FIG. 4, the main spring 28 is now providing a pop-up force to the main lever 30 in the direction of arrow 54, which will be applied to a striker of the hood 20 when the latch or latching mechanism 24 is released and the pop-up up system or mechanism 26 of the latch or latch mechanism 24 will be moved from the position illustrated in FIG. 4 (latch closed, pop-up up system or mechanism 26 ready to provide opening force) to the position illustrated in FIG. 2 (latch open, pop-up up system or mechanism 26 has provided opening force).

[0038] In other words, as the split lever 36 is rotated in the direction of arrow 54 from the position illustrated in FIG. 4, edge 76 of tab portion 58 of the split lever 36 contacts edge portion 78 of the main lever 30 in order to rotate the main lever 30 in the direction of arrow 54. The force for rotating the main lever 30 in the direction of arrow 54 is applied by spring 28.

[0039] Referring now to FIGS. 5-8, the latch or latch mechanism 24 with the pop-up up system or mechanism 26 in accordance with the present disclosure is illustrated. In FIGS. 5-8, other components (e.g., claw, pawl etc.) of the latch or latch mechanism 24 are illustrated, as well as the striker of the hood 20. FIGS. 5-7, generally correspond to position of the pop-up up system or mechanism 26 illustrated in FIGS. 2-4

[0040] In FIG. 5, the latch or latch mechanism **24** is in an open position and a claw **80** of the latch or latch mechanism **24** has not yet engaged a striker **82** of the hood. The claw **80** being rotatably mounted to the plate **32**. In FIG. 6, the cinch actuation of the latch or latch mechanism **24** has begun and the influence or biasing force of the main spring **28** is removed from the main lever **30** as previously discussed (e.g. first movement of movement of the cinch lever **42** in the direction of arrow **50** (FIG. 2 to FIG. 3), however, the main lever **30** and the claw **80** are still in their corresponding open positions. In FIG. 7, the striker **82** has been received in the claw **80** and the claw **80** has rotated in the direction of arrow **54** into a latch closed position (e.g., hood **20** is closed and the striker **82** is retained by the claw **80**). This may be facilitated by the cinching operation (e.g., second movement of the cinch lever **42** in the direction of arrow **50** (FIG. 3 to FIG. 4) as mentioned with respect to FIG. 4. During this movement the striker **82** moves in the direction of arrow **84** until the claw **80** reaches its closed or primary position (illustrated in FIG. 7). As mentioned above, movement of the main lever **30** from the open position to the closed position either by gravity and/or rotation of the claw **80** from the open position to the closed or primary position causes pawl **46** to no longer be in an engaged position such that end **70** of the pawl **46** no longer contacts tab portion **58** of the split lever **36** as mentioned above.

[0041] In this position a latch pawl **86** retains the claw **80** in the closed or primary position until the latch pawl **86** is moved from an engaged position (retaining the claw **80** in the closed or primary position—FIG. 7) to a disengaged position (allowing the claw **80** to rotate from the closed or primary position to the open position in the direction of arrow **56**—FIG. 6). Once this occurs, the claw **80** can move from the closed or primary position to the open position in the direction of arrow **56** while the main lever **30** provides a biasing or opening or pop-up force to the striker **82** in the direction of arrow **88** through release of the biasing force of the main spring **28** acting upon the split lever **36** which in turn moves the main lever **30** in the direction of arrow **54** via split lever **36**. In one non-limiting embodiment, the latch pawl **86** is rotatably mounted to the plate **32** for movement between at least the engaged position and the disengaged position.

[0042] Accordingly, and in accordance with the present disclosure, the pop-up system or mechanism is provided with a latch or latch mechanism that allows removal of the influence of the pop-up spring prior to a closing action of the hood **20** via a cinching operation that transitions the claw **80** from the open position to a closed or primary position. Additionally, a retention mechanism is provided that includes a pawl **46** component that keeps the main spring **28** disabled (e.g., prevented from providing a biasing force to the main lever **30**) until the latch or latch mechanism **24** is closed. This retention mechanism is then overridden or released when the latch or latch mechanism **24** reaches its fully closed position so that the main spring **28** can provide a pop-up or opening force when the latch **24** is released.

[0043] The term “about” is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, “about” can include a range of $\pm 8\%$ or 5% , or 2% of a given value.

[0044] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

[0045] While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential

scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A pop-up system (26) for a latch mechanism (24) of a vehicle, comprising: a main lever (30) pivotally mounted to a plate (32) of the latch mechanism (24); a split lever (36) pivotally mounted to the plate (32); a main spring (28) for providing a biasing force to the main lever (30) via the split lever (36); a bell crank (38) pivotally mounted to the main lever (30), the bell crank (38) being operably coupled to the split lever (36) at one end and a link (40) at another end, the link (40) pivotally mounted to a cinch lever (42), the cinch lever (42) being pivotally mounted to the plate (32); and a pawl (46) pivotally mounted to the plate (32), the pawl (46) being moveable between at least an engaged position wherein the pawl (46) maintains the split lever (36) in a first position wherein a biasing force of the main spring is prevented from being applied to the main lever (30) and a disengaged position wherein the biasing force of the main spring is applied to the main lever (30) in order to rotate the main lever into an open position when the latch mechanism (24) is released.
2. The pop-up system (26) as in claim 1, wherein the split lever (36) has a pin or protrusion (33) that is configured to contact a leg (35) of the main spring (28).
3. The pop-up system (26) as in claim 1, wherein the split lever (36) has a tab portion (58) configured to contact an end (70) of the pawl (46) when the pawl (46) is in the engaged position.
4. The pop-up system (26) as in claim 3, wherein a rotational force is applied to the main lever (30) by an edge (76) of the tab portion (58) of the split lever (36) contacting an edge portion (78) of the main lever (30).
5. The pop-up system (26) as in claim 1, wherein the bell crank (38) has a tab portion (53) configured to contact a tab portion (55) of the split lever (36) when the split lever (36) is in the first position.
6. The pop-up system (26) as in claim 1, wherein the main lever has a tab portion (72) configured to contact the pawl (46) and rotate the pawl (46) into the disengaged position.
7. The pop-up system (26) as in claim 1, wherein the latch mechanism (24) further comprises: a claw (80) rotatably mounted to the plate (32) and the cinch lever (42) is operably coupled to the claw (80).
8. The pop-up system (26) as in claim 7, wherein the latch mechanism (24) further comprises: a latch pawl (86) rotatably mounted to the plate (32) for movement between an engaged position and a disengaged position, the engaged position of the latch pawl (86) retaining the claw (80) in a closed position and the disengaged position of the latch pawl (86) allowing the claw (80) to rotate from the closed position to an open position.
9. The pop-up system (26) as in claim 8, wherein the split lever (36) has a pin or protrusion (33) that is configured to contact a leg (35) of the main spring (28).
10. The pop-up system (26) as in claim 9, wherein the split lever (36) has a tab portion (58) configured to contact an end (70) of the pawl (46) when the pawl (46) is in the engaged position.
11. The pop-up system (26) as in claim 10, wherein a rotational force is applied to the main lever (30) by an edge (76) of the tab portion (58) of the split lever (36) contacting an edge portion (78) of the main lever (30).
12. The pop-up system (26) as in claim 11, wherein the bell crank (38) has a tab portion (53) configured to contact a tab portion (55) of the split lever (36) when the split lever (36) is in the first position.
13. The pop-up system (26) as in claim 12, wherein the main lever (30) has a tab portion (72) configured to contact the pawl (46) and rotate the pawl (46) into the disengaged position.
14. The pop-up system (26) as in claim 13, wherein the latch mechanism is configured to retain a

vehicle hood (20) in a closed position by engaging a striker (82) of the vehicle hood (20).

15. The pop-up system (26) as in claim 1, wherein the latch mechanism (34) is configured to retain a vehicle hood (20) in a closed position by engaging a striker (82) of the vehicle hood.

16. A method for disengaging a main spring from a main lever of a pop-up system (26) for a latch mechanism (24) of a vehicle, comprising: pivotally mounting the main lever (30) to a plate (32) of the latch mechanism (24); pivotally mounting a split lever (36) to the plate (32); providing a biasing force by a main spring (28) to the main lever (30) via the split lever (36); pivotally mounting a bell crank (38) to the main lever (30), the bell crank (38) being operably coupled to the split lever (36) at one end and a link (40) at another end, the link (40) pivotally mounted to a cinch lever (42), the cinch lever (42) being pivotally mounted to the plate (32); and pivotally mounting a pawl (46) to the plate (32), the pawl (46) being moveable between at least an engaged position wherein the pawl (46) maintains the split lever (36) in a first position wherein a biasing force of the main spring is prevented from being applied to the main lever (30) and a disengaged position wherein the biasing force of the main spring is applied to the main lever (30) in order rotate the main lever into an open position when the latch mechanism (24) is released.
